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Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region

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Abstract: The project titled "Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region" The Rapti River, a vital water resource for the ecosystem and populace of the Gorakhpur region, faces severe environmental stress from escalating anthropogenic pressures. This proposal outlines a comprehensive study to analyze the spatio-temporal variations in the river's water quality. The primary objective is to create a detailed assessment of its hydro-chemical status, identify major pollution hotspots, and quantify the impact of key pollution sources .This research will involve systematic water sample collection from strategic locations, including upstream (baseline) sites, confluence points with major tributaries (like the Ami River), and areas downstream of significant discharge points (such as urban drains and the Gorakhpur Industrial Development Authority - GIDA). The analysis will focus on a suite of critical physico-chemical and biological parameters, including but not limited to: pH, Turbidity, Total Dissolved Solids (TDS), Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Hardness, nitrates, phosphates, and fecal coliform (MPN).The study hypothesizes that the river quality deteriorates significantly as it passes through the Gorakhpur urban agglomeration, with untreated domestic sewage and industrial effluents being the principal contributors. Particular attention will be given to the confluence with the heavily polluted Ami River, which is suspected to be a critical hotspot for chemical and heavy metal contamination.

***Keywords**: Physico-chemical parameters, pollution hotspots ,Biochemical Oxygen Demand(BOD), Chemical Oxygen Demand (COD),Dissolved Oxygen(DO).*